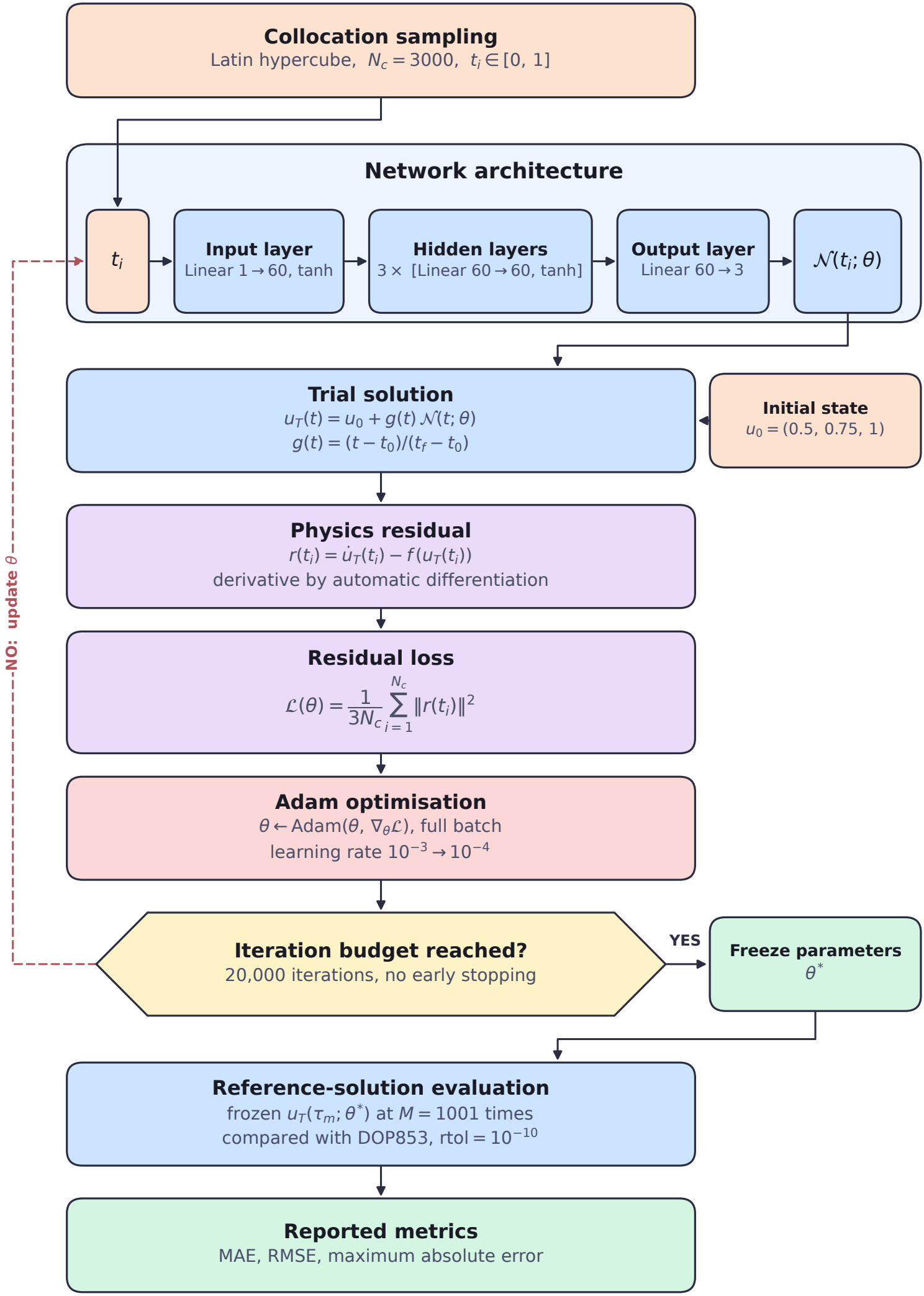
Collocation sampling
Latin hypercube, $N_c = 3000$, $t_i \in [0, 1]$
Network architecture
t_i
Input layer
Linear $1 \to 60$, tanh
Hidden layers
$3\times$ [Linear $60 \to 60$, tanh]
Output layer
Linear $60 \to 3$
$\mathcal{N}(t_i;\theta)$
Trial solution
$u_T(t) = u_0 + g(t)\,\mathcal{N}(t;\theta)$
$g(t) = (t - t_0)/(t_f - t_0)$
Initial state
$u_0 = (0.5, 0.75, 1)$
Physics residual
$r(t_i) = \dot{u}_T(t_i) - f(u_T(t_i))$
derivative by automatic differentiation
Residual loss
$\mathcal{L}(\theta) = \frac{1}{3N_c}\sum_{i=1}^{N_c} \|r(t_i)\|^2$
Adam optimisation
$\theta \leftarrow \mathrm{Adam}(\theta, \nabla_\theta \mathcal{L})$, full batch
learning rate $10^{-3} \to 10^{-4}$
Iteration budget reached?
20,000 iterations, no early stopping
NO: update θ
YES
Freeze parameters
θ^*
Reference-solution evaluation
frozen $u_T(\tau_m;\theta^*)$ at $M = 1001$ times
compared with DOP853, rtol $= 10^{-10}$
Reported metrics
MAE, RMSE, maximum absolute error